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Biology 2
0710

GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ADVANCED LEVEL**

Subject Title	**BIOLOGY**
Subject Code No.	**0710**
Paper No./Title	**Paper 2 Theory**

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Duration: Three Hours

Answer any FIVE out of the eight questions.

If more than five questions are answered, only the first five will be marked.

All questions carry equal marks. Marks allocated to parts of questions are indicated in brackets.

Illustrate your answers wherever desirable with large, clear and fully labelled diagrams.

You are reminded of the necessity for good English and orderly presentation in your answers.

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1. (a) What do you understand by:
   (i) Activation energy.
   (ii) Enzyme active site.
   (iii) Enzyme specificity.
   (b) Describe the different mechanisms that explain enzyme action.
   (c) Discuss the three factors affect that affect the rate of enzyme action.

(5, 5, 10 marks)
(Total = 20 marks)

2. (a) What is photoautotrophic nutrition?
   (b) Describe the events of the light independent reactions that occur in a monocot such as the sugar cane plant.
   (c) What are the advantages of these events in monocots over dicot plants?

(5, 10, 5 marks)
(Total = 20 marks)

3. (a) Define the following terms:
   (i) Filter feeding.
   (ii) Fluid feeding.
   (iii) Detrital feeding.
   (b) Outline the role of the following cells in digestion:
   (i) Parietal cells.
   (ii) Chief cells.
   (iii) Paneth cells.
   (iv) Goblet cells.

(8, 12 marks)
(Total = 20 marks)

4. (a) What is fermentation?
   (b) Why would a biotechnologist prefer the "Fed-Batch" process in fermentation?
   (c) With the aid of diagrams explain how genetic engineering is applied in the production of growth hormone (somatotrophin).

(3, 5, 12 marks)
(Total= 20 marks)

5. (a) Define biological conservation.
   (b) Why is the conservation of biodiversity necessary?
   (c) Describe the measures that can be used to efficiently conserve forests in Cameroon.
   (d) How may deforestation affect the environment?

(2, 4, 6, 8 marks)
(Total = 20 marks)

6. (a) Define the following terms as used in genetics:
   (i) Lethal gene.
   (ii) Codominance.
   (iii) Incomplete dominance.
   (b) The nature of the feather in birds is either normal or frizzled. Birds with homozygous dominant frizzled feathers die because of their inability to maintain their body temperature. Using genetic symbols and diagrams, show what will be the outcomes between two frizzled feathered birds.
   (c) Why do the above results deviate from the normal Mendelian ratio?

(9, 9, 2 marks)
(Total = 20 marks)

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7. (a) What do you understand by tissue respiration?
   (b) (i) What are the major phases of tissue respiration.
   (ii) Where do these major phases occur?
   (iii) What are the respective products of each of the major phases of tissue respiration?
   (iv) What is the net ATP produced in each phase of tissue respiration?
   (c) What is the fate of pyruvate in anaerobic respiration?

(3, 12, 5 marks)
(Total = 20 marks)

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8. (a) (i) Define the term homeostasis.
   (ii) Explain the process of negative feedback in living organisms.
   (b) Describe how insulin and glucagon regulate the blood glucose concentration in mammals.

(4, 4, 12, marks)
(Total = 20 marks)

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